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Amalgam Based Multilingual Text Summarization For Devanagari Languages

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***Abstract*—The absence of efficient text summarizers for languages based on the Devanagari script like Hindi, Marathi and Sanskrit exhibit a conspicuous void in field of literature being devoid of the linguistic masterpieces. This study introduces a customized Multilingual Text Summarizer for Marathi and Hindi that makes use of a combination Python's TextRank algorithm and Bag of Words (BOW) algorithm alongside extractive summarization techniques using a flask web application. By preserving the meaning and extracting most used words and sentences, our approach yields accurate summaries. The dataset used contains news articles and their summaries for testing our model. Empirical assessments using ROUGE approach confirms its efficiency in generating effective summaries with over 88% score.**

***Index Terms*—Bag of Words, Devanagari, extractive, multilingual, ROUGE, summarization, TextRank**

I. Introduction

In the current era, the digital world grapples with an overwhelming influx of textual data across various languages. This calls for the pressing challenge of efficiently processing and comprehending vast volumes of information. As a solution, text summarization technology has come into picture, aiming to transform lengthy texts into concise, coherent summaries without sacrificing crucial content. However, there is still a lack of effective summarizers for regional languages particularly concerning Devanagari-based languages such as Hindi, Marathi and Sanskrit.

As the demand for effective text summarization tools for Devanagari languages has become the need of the hour, as we are bombarded with information from all kinds of media predominantly available in English. These languages serve as media of communication, education, and entertainment for millions. However, a dearth of summarization tools withholding the sentiment with which text is written is evident. Hindi and Marathi collectively have over 600 million population calling them their mother tongue. Thus, it has become a necessity to effectively summarize the available texts and rich literature. With its ancient roots and deep cultural impact, both Hindi and Marathi need a text summarizer preserving the text quality without literal translation of every word into English and generating summary to be translated back to desired language.

The objective of this paper is to propose a model that implements Python's TextRank algorithm, Bag of Words algorithm [15] and utilizes extractive summarization techniques to enhance the scope of Devanagari based languages to reach a greater audience. In extractive summarization every sentence is ranked in each data based on the meaning and relevance to the entire textual data and then combine the sentences with highest ranks to form the summary. Tokenization of sentences makes generation of extractive summary easier, and significantly enhances model accuracy. ROUGE [16] (or Recall Oriented Understudy of Gisting Evaluation) is implemented to certify efficiency which comes out to be over 88% as compared to around 86-87% in other models. The combination of BOW and TextRank is the reason for this improvement in results. The remainder of this paper is organized as follows: Section 2 provides a comprehensive literature review, aiming to pinpoint gaps and limitations in current research within this domain. In Section 3, we present our proposed model, complete with the algorithms employed, pseudo code, and experimental implementation and 5 are dedicated to showcasing the results obtained from our model and conducting a comparative analysis with existing models.

II. Literature Review

The model [1] extracts 11 features from the text such as Term Frequency-Inverse Sentence Frequency, Similarity between sentences, Numerical Data, Location of sentence [7] in the document, etc which are then given as input to a Fuzzy Logic Model to rate the sentences from 0-1 based on

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increasing importance. An edge value is then set to summarize the text. This research introduces a Multilingual Document Summarization (MDS) [13] model to address cross-lingual summarization challenges. The model uses mathematical formulae and pre-processing steps to pull features like sentence position, length, and sectional words. A CNN classifier is then deployed to filter sentences, followed by TextRank-based scoring [14]. This combination of supervised-unsupervised approach shape up for effective multilingual summarization.

Prior to the processing phase, Marathi text is structured methodically and then during Processing, diverse factors which influence sentence importance [2], like Marathi keyword recognition, relative sentence length, and numerical data evaluation, are determined. Linguistic features for significant sentence selection include recognizing Marathi headlines and neighbouring lines. The review [10] [11] consists of diverse summarization methods, covering both abstractive and extractive approaches. It dives into structured and semantic-based methods, featuring algorithms such as TextRank, genetic algorithms, and deep learning.

The research [3] anticipates progress in multilingual models, fine-tuned summarization, domain acclimation, user preferences, multi-document summarization, and wholesome evaluation metrics as areas for future research. The ultimate objective is to enhance information summarization across languages to promote accessibility and comprehensiveness. In [5] [6] Indian languages employ methods like sentence scoring, ROUGE evaluation, and subgraph analysis. Notable contributions include the development of a Rich Semantic Graph [12] for Marathi, language-independent approaches, and action word recognition in Marathi.

III. Model and Methods

In this model, a combination of the Bag of Words (BoW) model with the TextRank algorithm is proposed for a more powerful approach towards text summarization and information extraction tasks. This combination leverages the strengths of both techniques to produce more informative and contextually relevant summaries. Text summarization aims to condense a large body of text into a shorter, coherent, and informative summary. The Bag of Words (BoW) model is a fundamental technique for text representation and information retrieval. It represents a document as a collection of word frequency vectors, disregarding word order and structure. On the other hand, TextRank is an extractive summarization algorithm works on identification of importance of sentences in a document based on their relationships in a sentence similarity graph. Thus, BoW provides a solid foundation for word-level representation and frequency-based importance, while TextRank identifies relationships between sentences and ranks them accordingly. This hybrid approach allows for more contextually relevant summaries, by capturing both word-level and sentence-level importance, resulting in more informative and contextually relevant summaries.

In the given figure, the design of proposed model through a diagrammatic approach is explained. As the user provides the desired input text to be summarised to this model, it would undergo through a preprocessing. The corpus would be cleaned and then feature extraction for the data would be done. Bag of Words is applied to words without taking the order or structure of sentence for choosing the words. Furthermore, TextRank is used to enhance the summarisation process by following a graphical approach to classify sentences with respect to their importance and generate a summary as per their ranking.

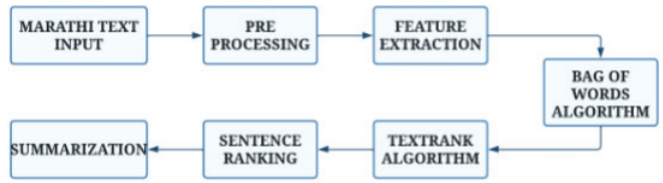

Fig. 1. Design of Module

A. Bag Of Words (BOW) Algorithm

The Bag of Words (BoW) model is a very popular natural language processing (NLP) technique for text analysis. It represents text data as a set of words, ignoring the order and structure of the sentences. BOW tokenises the entire document into short sentences or words. These tokens are further used to build a vocabulary as $V = v_1, v_2, ..., v_m$, where m is the sum of unique words in the thesaurus. A vector is created for each text document in corpus.The BoW vector is basically a vector of word frequencies, where each element of the vector corresponds to a word in the vocabulary.

$$freq(w_i, D_j) = X_o \tag{1}$$

where, X_o is No. of times word w_i appears in document D_j

BoW(D_j) = $[freq(v_1, D_j), freq(v_2, D_j), ..., freq(v_m, D_j)]$
where each element of the BoW vector for a document Dj, BoW(Dj), is a count of the word frequencies from the vocabulary.

If there is no occurence of the word in text document, its frequency is set to 0. A DTM or Document Term Matrix is used when multiple text documents are used. Here, each row represents the document while column represents the vocabulary. Each element of DTM is word frequency.
DTM =$[BoW(D_1), BoW(D_2), ...BoW(D_n)]$.

Document Term Matrix is mostly a sparse matrix because most elements are zeros since not all words from the vocabulary appear in every document.

B. PageRank Algorithm

Google Search implements the PageRank Algorithm, named after their founder Larry Page, for ranking websites and optimising search engine outputs. It is a way of measuring the importance webpages being displayed to users. PageRank

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calculates a rough estimate of the importance of a website by counting the number and quality of links to that page. The hidden supposition that will be that more significant sites are probably going to get additional connections from different sites.

Similar to most algorithms, the entire text document is tokenized into smaller sentences or words. D = $S_1, S_2, ..., S_n$ Each token (sentence or word) becomes a node in a graph $G(V, E)$, where V depicts the set of nodes (sentences or words) and E depicts the group of edges connecting these nodes. In PageRank, measure of similarity between nodes S_i and S_j is denoted as sim(S_i, S_j). The weights to the edges are assigned based on these similarity scores: w(S_i, S_j) = sim(S_i, S_j). The graph is fully connected, so there is an edge between every pair of nodes with weights w(S_i, S_j). Higher similarity leads to stronger connections (higher weights) between nodes. Common choices include cosine similarity, Jaccard similarity, or even simple overlap of words.

$$PR(S_i) = (1 - d) + d * \sum(PR(S_j) * w(S_j, S_i)) \quad (2)$$

where:

$PR(S_i)$ is the PageRank score of node S_i.

d is a damping factor, typically set between 0.85 and 0.95, to balance exploration and exploitation.

$\sum$ represents the summation over all nodes Sj that have an edge to Si.

The PageRank scores are iteratively updated until the scores no longer change significantly. Finally, the final PageRank scores for all nodes (sentences or words) in the graph are obtained. Higher PageRank scores indicate greater importance. Thus, for summarization, the higher ranked words, or text is extracted based on above score. Either a threshold score is set to select nodes with scores above a certain value or the top N nodes with the highest PageRank scores are chosen.

C. TextRank Algorithm

Text rank algorithm which is based on Google's PageRank is now become the significant graph-based technique in the realm of NLP (Natural Language Processing) as well as text summarization. It was first introduced by Mihalcea and Tarau in 2004, and has gained popularity in text mining over the years.

Tokenization the process of dividing a text into multiple individual words or tokens. It's a very crucial step in context to TextRank, as it breaks the entire text into smaller meaningful units and further forms the foundation of comparing the content.

$$T = W_1, W_2, W_3, ..., W_N$$

In the next sentence splitting step, the input text is divided into individual sentences. Sentences are the building block of any text document. By splitting these sentences, we create discrete units that can be ranked for importance.

$$T = S_1, S_2, S_3, ..., S_M$$

All sentences provided as input text are treated as nodes in a graph and similarity between nodes(sentences) are known as edges. Cosine Similarity is used for measuring the similarity between two sentences and it's computed by the cosine of the angle between two vectors, which represents the sentences. Higher the Cosine Similarity, strongest the connection between the sentences. Therefore, this structure of graph helps TextRank to capture the relation between sentences and helps in identifying which sentences are most important in the context of input text. TextRank uses a similarity matrix often denoted by "S", to calculate and store the pairwise similarity scores between sentences.

$$Similarity(A, B) = \frac{A \cdot B}{\mid\mid A \mid\mid \cdot \mid\mid B \mid\mid} \quad (3)$$

where, A and B are two sentence vector,
$A \cdot B$ represents dot product of A and B,
$\mid\mid A \mid\mid$ and $\mid\mid B \mid\mid$ represents Euclidean norms (magnitudes) of A and B.

This formula yields a value between -1 and 1, with higher values indicating greater similarity. TextRank has made a remarkable impact in various natural language processing tasks, and provided the much needed impetus in extractive summarization. TextRank uses a graph-based ranking algorithm for assigning importance scores to every sentence in the graph. Initial score for each sentence is set to 1 and these scores represent the importance of each sentence in isolation then iteratively these scores gets updated based on the scores of sentences connected to them.

$$Score(S[i]) = (1 - d) + d * \sum(\frac{S[j]}{OutDegree[j]}) \quad (4)$$

where, Score(S[i]) is updated score of sentence i,
d is Damping factor which is usually around 0.85,
S[j] is score of sentence j,
OutDegree[j] is Out-degree of sentence j, which represents the number of sentences connected to it.

This iteration continues till scores of sentences converge which means that the scores no longer change significantly from one iteration to next. After Convergence, each sentence are ranked based on their final scores, the top-ranked sentences are selected to create summary for the input text, based on a threshold value.

D. Frontend

A Flask web application serves as the frontend for this project. The structure and layout of the web pages is defined using HTML. The UI contains an input text area where the user will input the textual data to be summarized. CSS is used to make the web page look visually appealing.

E. Psuedo Code

- *Start*
- *Preprocessing*
  - *cleaned_text = clean_text(input_text)*

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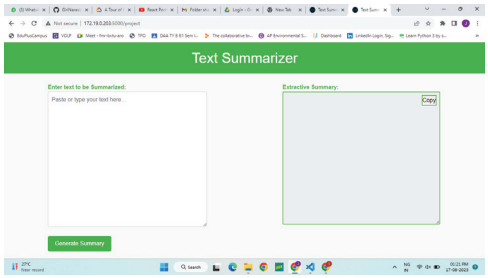

Fig. 2. User Interface

  - *sentences = tokenize_into_sentences(cleaned_text)*
- *Calculate Similarity Matrix*
  - *Vectorize the sentences using a suitable method*
- *TextRank-based Summarization*
  - *graph = create_similarity_graph(similarity_matrix)*
  - *scores = calculate_text_rank_scores(graph)*
  - *summary = select_top_sentences(ranked_sentences)*
- *Return the generated summary*
- *End*

F. Flowchart

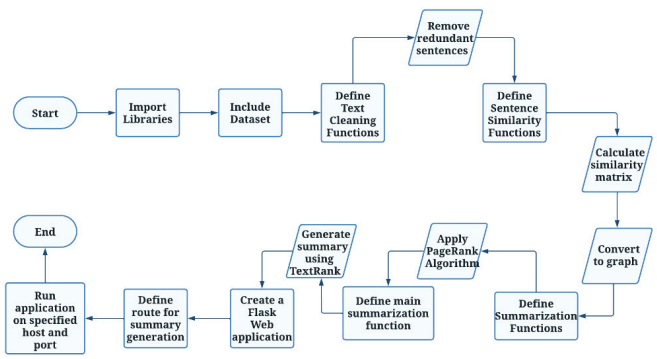

Fig. 3. Flowchart

IV. Performance Evaluation

ROUGE or Recall-Oriented Understudy for Gisting Evaluation is an evaluation metric for comparing the machine generated summary to reference (human-generated) text. ROUGE mostly focuses on extractive summarization by measuring the overlap of n-grams (contiguous sequences of words) and other text units between the system-generated output and the reference text. Precision measures the number of n-grams in the system-generated output appearing in the reference text.

$$P_n = \frac{n-grams \quad ref \cap \quad grams \quad sys}{Total \quad n-grams \quad sys} \tag{5}$$

Recall measures the number of the n-grams covered in the reference text by the system-generated output.

$$Rn = \frac{n-grams \quad ref \cap \quad n-grams \quad sys}{Total \quad n-grams \quad ref} \tag{6}$$

The F1 score is the harmonic mean of precision and recall and can be used to balance trade-off.

$$F1_n = \frac{2 * P_n * R_n}{P_n + R_n} \tag{7}$$

ROUGE is calculated for for multiple values of ”n” and then mean of it is taken.

$$ROUGE = \frac{1}{N} * \sum ROUGE - n \tag{8}$$

N is the number of different values of ”n” considered (e.g., uni-grams, bi-grams, etc.), and $\sum$ represents the summation over these values. In addition to n-grams, using ROUGE-L i.e. he longest common sub-sequence (LCS) or the longest contiguous sequence of words appearing between both the system-generated output and the reference . Also, ROUGE-W or weighted ROUGE assigns different weights to different n-grams based on their positions in the text, to give different importance to the n-grams appearing at the beginning and ending of text.

V. Results and Discussions

TextRank Algorithm is very powerful tool for Extractive Summarization. It does not require any training data and can be applied to various languages and domains. It is a valuable tool for automatic summarization as it ranks all sentences based on their relation with other sentences in input text. A combination of Bag Of Word (BOW) and TextRank Algorithm is implemented to obtain best Rouge scores which is an empirical test for machine generated summaries when compared with human generated summaries. This combination allows for the extraction of key information from the document, as represented by BoW, and the identification of important sentences based on their content and relationships, as determined by TextRank. It results in an effective and data-driven extractive text summarization technique.

Example:
Input Text:

Summary Generated:

In text summarization, ROUGE-1 evaluates unigrams (single words) for overlap between the generated and reference text. ROUGE-2 extends this to bigrams (pairs of adjacent words), considering word pairs' cohesiveness. ROUGE-L, meanwhile, focuses on the longest common subsequence, accommodating word order variations in summarization. These variants cater to different aspects of summarization quality assessment, from individual words to word pairs and content reordering.

The following ROUGE scores are achieved through thorough analysis and pruning mentioned above in the Table 1. The generated ROUGE scores are marginally better than all existing models available [10] [11] [16]. And the same has been mentioned graphically in Fig. 4.

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TABLE I
ROUGE SCORES FOR DIFFERENT SUMMARIZATION MODELS/METHODS

Model	Rouge-1	Rouge-2	Rouge-L
Our Model	0.88014	0.87327	0.80124
TF ISF	0.67	0.62	0.6
Bag of Words	0.8684	0.8602	-
Text Rank	0.8004	0.7909	-
TF IDF	0.747	0.741	-
IndicBART	0.6104	0.515	0.475
mT5-small	0.595	0.49	0.46
Simple Extractive	0.834	0.747	-
T5	0.2148	0.2021	-
Unsupervised	0.45952	0.19338	0.21763
Wikipedia-based	0.46	0.23	-
Deep Neural Network	0.327	0.122	0.29
Distributional Semantics	0.33	0.07	0.2

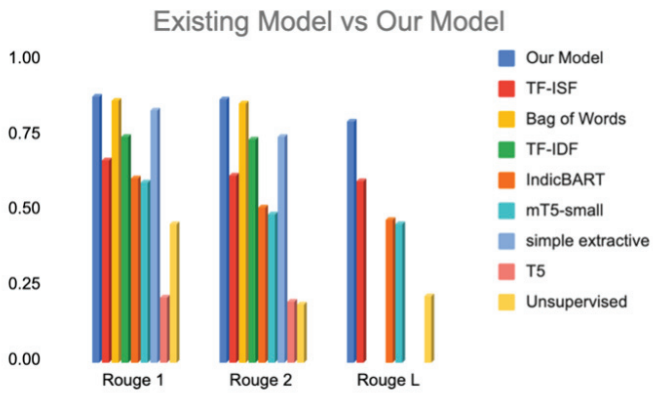

Fig. 4. Result Comparison with existing model

VI. CONCLUSION

The need of automatic text summarization has exponentially risen as copious amounts of data is transferred and processed each second. This paper highlights need of text summariser for regional languages and proposes a model for such regional languages of India, Hindi, and Marathi. This text summarization will not just provide impetus to regional languages to exhibit their literature to a wider audience, but also simplify the texts for existing user base. An extractive approach is proposed for texts summarization to include feature extraction, sentence selection, tokenization based on graphical methods.

This model is useful in various domains and spheres of life ranging from education, journalism, research, restoring literal heritage and recreation. The paper throws a light on various Natural Language Processing (NLP) techniques including text lemmatisation, tokenisation, stemming, etc. Bag of Words (BOW) is used for text modelling whereas, TextRank Algorithm, based on PageRank. is used for keywords extraction, sentence ranking and text summarization purpose. ROUGE (Recall Oriented Understudy for Gisting Evaluation) is an assessment metric working on principle of comparing machine generated summary to any (Ideal) summaries written by a human. It has been observed that this model has performed comparatively well than all existing solutions. The proposed model can be enhanced in future by extending to abstractive summarization. Furthermore, inclusion of multiple languages written in same script can be supported by modifying the stopwords. It can also be extended to incorporate the speech to text to aid the people with special needs and also be more user friendly.

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